

PIERRE DELIGNE · D030

Editorial Apparatus

Authority · topology · exclusions · textual audit

This apparatus binds the 34 physical authority pages to printed pages 313–346, records copy-matter exclusions, distinguishes the comparison witness, and identifies the embedded source-English Koblitiz–Ogus appendix. It does not silently amend either language edition.

Controlling identities

Authority. IAS scan, 34 pages; SHA-256 6CFACB0F8742314FB9FAEB3585159BDC13C3FF6E93686E3FE0A23AD775E8B574.

Comparator. Collected-paper witness, comparison only, 34 pages; SHA-256 30C125672931928F18E6CBC1930DB1BC2920DDA1685A68C3578F2D8E2B9158A4.

Prior work. Exact inherited ZERO_ACCEPTED carrier; SHA-256 7EDC48FE09334A876C29DB564C067721BFA0F813EC21FA52D1A7929CBF9A6D98.

Image fallback. Lossless authority crop 4DB161893339B5781D6A7F1636A4C682354CB06A5A50912688538AB037978DA9; presentation derivative 222B3D1A4B351A59C88B1829F75D197616078965EB053F30BCB847B0908C7818.

Copy-matter rule

Repeated proceedings headers, repeated folios/bylines, scan-edge black bands, neighboring-article slivers, and terminal blank remainder are excluded from the reader text. Publication citation, classifications, and copyright notice are retained here. The article proper and the source-English appendix are included in full.

Topology overview

Physical 1 · printed 313 · P01 FRENCH MAIN BODY. printed 313; proceedings citation; exact title; P. DELIGNE byline; introduction; §0 Motifs; subject classifications and copyright

Physical 2 · printed 314 · P01 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 3 · printed 315 · P01 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 4 · printed 316 · P01 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 5 · printed 317 · P01 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 6 · printed 318 · P01 FRENCH MAIN BODY. printed 318; §1 Énoncé de la conjecture (cas rationnel)

Physical 7 · printed 319 · P02 FRENCH MAIN BODY. printed 319; Definition 1.3 continuation; source diagram (1.3.1); Proposition 1.4 and proof

Physical 8 · printed 320 · P02 FRENCH MAIN BODY. printed 320; §2 Énoncé de la conjecture (cas général)

Physical 9 · printed 321 · P02 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 10 · printed 322 · P02 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 11 · printed 323 · P02 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 12 · printed 324 · P02 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 13 · printed 325 · P03 FRENCH MAIN BODY. printed 325; terminal determinant-line paragraph of §2; §3 Exemple: la fonction ζ

Physical 14 · printed 326 · P03 FRENCH MAIN BODY. printed 326; §4 Compatibilité à la conjecture de Birch et Swinnerton-Dyer

Physical 15 · printed 327 · P03 FRENCH MAIN BODY. printed 327; §5 Compatibilité à l'équation fonctionnelle

Physical 16 · printed 328 · P03 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 17 · printed 329 · P03 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 18 · printed 330 · P03 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 19 · printed 331 · P04 FRENCH MAIN BODY. printed 331; §6 Exemple: les fonctions L d'Artin

Physical 20 · printed 332 · P04 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 21 · printed 333 · P04 FRENCH MAIN BODY. printed 333; §7 Fonctions L attachées aux formes modulaires

Physical 22 · printed 334 · P04 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 23 · printed 335 · P04 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 24 · printed 336 · P04 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 25 · printed 337 · P05 FRENCH MAIN BODY. printed 337; §8 Caractères de Hecke algébriques

Physical 26 · printed 338 · P05 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 27 · printed 339 · P05 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 28 · printed 340 · P05 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 29 · printed 341 · P05 FRENCH MAIN BODY. continuous source-native text, mathematics, and layout

Physical 30 · printed 342 · P05 FRENCH MAIN BODY. printed 342; French main body closes before appendix boundary

Physical 31 · printed 343 · P06 SOURCE ENGLISH KOBLITZ OGUS APPENDIX. printed 343; source-English Appendix by N. Koblitz and A. Ogus begins; exact appendix title and definitions

Physical 32 · printed 344 · P06 SOURCE ENGLISH KOBLITZ OGUS APPENDIX. continuous source-native text, mathematics, and layout

Physical 33 · printed 345 · P06 SOURCE ENGLISH KOBLITZ OGUS APPENDIX. printed 345; source-English appendix theorem/proof; Bibliographie begins, entries 1-5

Physical 34 · printed 346 · P06 BIBLIOGRAPHY TERMINAL ADDRESS MIXED NEUTRAL. printed 346; bibliography entries 6-19; SGA references; IHÉS address; blank remainder; next-article sliver excluded; EOF

Page-by-page apparatus

Authority physical 1 · printed 313

Zone: FRENCH MAIN BODY

Printed page 313; French main body. The diplomatic source includes the exact title, Deligne byline, opening, §0, and the beginning of 0.1. The proceedings masthead, folio, and footline are copy matter rather than article text; recorded here: Proceedings of Symposia in Pure Mathematics, Vol. 33 (1979), part 2, pp. 313–346; AMS (MOS) subject classifications (1970), Primary 10D25, 14K10; © 1979, American Mathematical Society. Authority replay fixes L_3/L_4 , “dù à Grothendieck,” and the printed L_k/Sym^k notation. The inherited candidate was consulted only after source freeze and remains locator-only.

Authority physical 2 · printed 314

Zone: FRENCH MAIN BODY

Printed page 314; French main body, beginning with the continuation “lisée” from page 313. The running folio/byline, scan-edge black band, and neighboring-page sliver are excluded. Authority notation $\mathcal{M}_{\mathrm{eff}}(k)$, $\Pi_{k'}/k$, and the functor F is retained. No image fallback is required.

Authority physical 3 · printed 315

Zone: FRENCH MAIN BODY

Printed page 315; French main body, completing 0.2.4 and continuing through 0.4. Authority uses F_∞ in 0.2.5, the finite-adelic product, and the two comparison isomorphisms; these were checked formula by formula. Running header/folio and scan-edge material are excluded.

Authority physical 4 · printed 316

Zone: FRENCH MAIN BODY

Printed page 316; French main body. Diagram (0.5.4) is represented natively as a commutative text diagram, so no crop is used. Authority notation $Z_R^d(X)$ and $Z_R^{\dim(Y)}(X \times Y)$ is retained. Running folio/byline, scan-edge band, and neighboring-page sliver are excluded.

Authority physical 5 · printed 317

Zone: FRENCH MAIN BODY

Printed page 317; French main body. Proposition 0.8, Espoir 0.10, and the added-in-proof paragraph are replayed. The authority product $H^{2d}(X, k \times \mathbb{A}^f)(d) = H_{\mathrm{DR}}^{2d}(X)(d) \times H^{2d}(X, \mathbb{A}^f)(d)$ and the subset chain in 0.8(ii) are preserved. Running header/folio and scan-edge material are excluded.

Authority physical 6 · printed 318

Zone: FRENCH MAIN BODY

Printed page 318; French main body. The accented heading “Énoncé de la conjecture (cas rationnel)” is preserved. Equations (1.2.1)–(1.2.3) were checked against the authority; (1.2.3) has Λ on both sides. The page ends at the printed hyphen in “multiplica-”. Running folio/byline, scan-edge band, and neighboring-page sliver are excluded.

Authority physical 7 · printed 319

Zone: FRENCH MAIN BODY

Printed page 319; French main body. The page completes Definition 1.3, reproduces diagram (1.3.1), and states Proposition 1.4. The diagram is supplied as an untouched pixel crop from the attached authority render and a separately hashed presentation derivative; its upper-left and lower-right hatching, upper-right minus sequence, lower-left plus sequence, axes, and “ $(-1, -1)+$ ” annotation were checked visually. Running header/folio, scan-edge band, and neighboring-page sliver are excluded.

Authority physical 8 · printed 320

Zone: FRENCH MAIN BODY

Printed page 320; French main body. Sections 1.5–1.7, equations (1.7.1)–(1.7.3), Conjecture 1.8, and the opening of §2 were replayed. The authority quotient-space notation $H_{\overline{DR}}^{\pm}$ and the accented heading “Énoncé de la conjecture (cas général)” control; the inherited unaccented heading is not promoted. Running folio/byline and scan-edge material are excluded.

Authority physical 9 · printed 321

Zone: FRENCH MAIN BODY

Printed page 321; French main body. The two equivalent constructions of motives with coefficients, the tensor/dual formulas, the trace pairing, and extension/restriction of coefficients were checked against the authority. The inherited candidate was consulted only after freeze and remains locator-only. Running header/folio and scan-edge material are excluded.

Authority physical 10 · printed 322

Zone: FRENCH MAIN BODY

Printed page 322; French main body. Equations (2.2.1)–(2.2.3), the $E \otimes \mathbb{C}$ decomposition map, Proposition-Definition 2.3, and §2.4 were checked formula by formula. In condition (ii), the authority has $\tilde{M}(1)$, which is retained. Running folio/byline and scan-edge material are excluded.

Authority physical 11 · printed 323

Zone: FRENCH MAIN BODY

Printed page 323; French main body. The Hodge decomposition, Proposition 2.5, equations (2.6.1)–(2.6.2), period determinants, and Conjectures 2.7–2.8 were replayed. The authority order of I and $I^{\pm}$ and its quotient-space notation $H_{\overline{DR}}^{\pm}$ are retained. Running header/folio and scan-edge material are excluded.

Authority physical 12 · printed 324

Zone: FRENCH MAIN BODY

Printed page 324; French main body. Remarks 2.9–2.12 and the beginning of 2.13 were replayed, including the reduced determinant, the authority equation $\iota_C L^*(M, s) = L^*(e(M \otimes_E F), s)$, and $W_{F,L}$. The page ends at the displayed $D \otimes F$ -module isomorphism. Running folio/byline and scan-edge material are excluded.

Authority physical 13 · printed 325

Zone: FRENCH MAIN BODY

Printed page 325; French main body. The page completes §2.13 and begins §3. Authority replay fixes the determinant-line object δ in $\delta(H_B^+(M), H_{\overline{DR}}^+(M))$, with $\det \text{red}(I^+) \in \delta \otimes \mathbb{C}$ and e a basis of δ . It also preserves $W_{F,L}$, $H_1(G_m)$, the plain $\mathbb{Z}$ in the Hodge realization, c^e in (3.1.3), and the page-ending continuation of §3.2. Running header/folio, scan-edge band, and neighboring-page sliver are excluded.

Authority physical 14 · printed 326

Zone: FRENCH MAIN BODY

Printed page 326; French main body. The accented heading “Compatibilité à la conjecture de Birch et Swinnerton-Dyer” is retained. Equations and periods in §4.1 were checked against the authority, including the printed c_i/e_i notation change and $\int_{A(\mathbb{R})} |\omega| = [A(\mathbb{R}):A(\mathbb{R})^\circ] \int_{A(\mathbb{R})^\circ} |\omega|$. The page ends with the continuation “propre et”. Running folio/byline and scan-edge material are excluded.

Authority physical 15 · printed 327

Zone: FRENCH MAIN BODY

Printed page 327; French main body. The page completes §4.3 and begins the accented heading “Compatibilité à l’équation fonctionnelle.” The exact cohomology sequence, Proposition 5.1, determinant-line identities (5.1.1)–(5.1.4), and Lemmas 5.1.5–5.1.6 were replayed. In particular, (5.1.4) has a plain inverse on $\delta(G, X/F)$, and Lemma 5.1.6 uses the induced map f_G^{-1} . Running header/folio and scan-edge material are excluded.

Authority physical 16 · printed 328

Zone: FRENCH MAIN BODY

Printed page 328; French main body. Equations (5.1.7)–(5.1.9) and the opening of §5.2 were checked formula by formula. The source retains $c^{\pm(-1)^n}(M(n))$, the exponent $-d^{\pm}(M) \cdot n$, the enlarged Weil group W' , and the authority functional equation with Λ . The page ends at the continuation “à associer”. Running folio/byline and scan-edge material are excluded.

Authority physical 17 · printed 329

Zone: FRENCH MAIN BODY

Printed page 329; French main body. The page completes the local ε -factor construction, Remark 5.2.1, and the archimedean table in §5.3. The native text table preserves $\Gamma_{\mathbb{C}}(s-p)$ for both the complex row and the real two-dimensional row, and Proposition 5.4 uses powers of 2π rather than $2\pi i$. No image fallback is required. Running header/folio and scan-edge material are excluded.

Authority physical 18 · printed 330

Zone: FRENCH MAIN BODY

Printed page 330; French main body. The page completes Proposition 5.4, gives (5.4.1), the contribution table, Proposition 5.5, equations (5.5.1)–(5.5.3), and begins Theorem 5.6. The authority contribution table has $F_{\infty} = -1$ in both (p,p) rows, and $\det M = \wedge^{d(M)} M$. The page ends before the theorem’s displayed formula on printed page 331. Running folio/byline and scan-edge material are excluded.

Authority physical 19 · printed 331

Zone: FRENCH MAIN BODY

Printed page 331; French main body. The page completes Theorem 5.6 and opens §6 on Artin L-functions. Authority replay fixes the theorem’s $2\pi/2\pi i$ distinction, the exponent and i^{ε} factor in (5.6.1), and the contravariant direction $F^*: \mathbb{Q}^{Y(\mathbb{Q})} \rightarrow \mathbb{Q}^{X(\mathbb{Q})}$. The definition of Artin motives, the Galois equivalence, and the matrix-transpose correspondence are retained as native text. No image fallback is required. Running heading, folio, scan-edge band, and neighboring-page sliver are excluded.

Authority physical 20 · printed 332

Zone: FRENCH MAIN BODY

Printed page 332; French main body. The page gives the Betti, l-adic, and de Rham realizations of Artin motives, the Gauss-sum period, Proposition 6.5, and Conjecture 6.6. Authority pixels control the material formula $\varepsilon^*(D) \sim (2\pi)^{-n} i^{\eta-n} \delta(D)$, the f^{ε} factor in the functional-equation constant, and $\delta(D) = (2\pi i)^n \delta([\varepsilon])$; these differ from the inherited candidate. All formulas are rendered natively; no image fallback is used. Running byline/folio and scan-edge material are excluded.

Authority physical 21 · printed 333

Zone: FRENCH MAIN BODY

Printed page 333; French main body. The page proves Proposition 6.7 and begins §7. Authority replay preserves $n \leq 0$ and $\tau L(\sigma, V(n)) = L(\tau\sigma, V(n))$, then freezes the mandatory F11 heading exactly as “Fonctions L attachées aux formes

modulaires.” The modular-form Euler factor, determinant $[\varepsilon^{-1}](1-k)$, $\Gamma_0(N)$ topology, and starred Hecke operator T_n^* in (7.2.1) are retained. No image fallback is required. Running header/folio and scan-edge material are excluded.

Authority physical 22 · printed 334

Zone: FRENCH MAIN BODY

Printed page 334; French main body. The page continues §7.2 through §7.4. Authority replay preserves $P(T_n - a_n \Delta)^*$, the compact-support map (7.2.2), the minus sign in $L(M(f), 1) = -\int_0^{i\infty} \omega_f$, both terms of Remark 7.3, and the trans-
pose/star distinction in the $\Gamma_1(N)$ Hecke action. The period system $(\int_0^{i\infty} \omega_{\sigma f})_{\sigma}$ is frozen from the scan. No image fallback is used; running byline/folio and scan-edge material are excluded.

Authority physical 23 · printed 335

Zone: FRENCH MAIN BODY

Printed page 335; French main body. The page completes the modular-period argument, gives Remark 7.5 and §7.6, and states Proposition 7.7. Authority replay fixes the reference to 2.8, the $H_c^1 \rightarrow H^1$ topology, the universal-elliptic-curve coefficient system, the Eichler integrals, the odd/even period exponents, and the off-diagonal matrix of F_{∞} . The inherited candidate changes several of these structures and remains locator-only. No image fallback is required. Running heading/folio and scan-edge material are excluded.

Authority physical 24 · printed 336

Zone: FRENCH MAIN BODY

Printed page 336; French main body. The page proves Proposition 7.7 by wedge-pairing and degree counts, then begins §7.8 with (7.8.1)–(7.8.3). The wedge structures, odd/even degree tables, local Euler factor, Gauss-sum determinant period, and sign rule $\pm = (-1)^m$ are retained as native preformatted text. No image fallback is used. Running byline/folio and scan-edge material are excluded.

Authority physical 25 · printed 337

Zone: FRENCH MAIN BODY

Printed page 337; French main body. The page completes §7.8 with (7.8.4), the symmetric-power local factor and critical-value formula, then opens §8. The accented heading “Caractères de Hecke algébriques” is the mandatory F12 replay gate. Conjecture 8.1 is retained with rank one over k , the K_i -decomposition, Hodge filtration, and equations (8.1.1)–(8.2.1). No image fallback is used. The running header/folio, black scan edge, and neighboring-page sliver are excluded.

Authority physical 26 · printed 338

Zone: FRENCH MAIN BODY

Printed page 338; French main body. The page continues Remark 8.2, gives Examples 8.3–8.5, defines the period $p'(\chi)$, and begins its CM interpretation. The authority’s inverse determinant in Example 8.3, the $k \otimes E$ and E^{*S} quotient, the integral $\int_Z \omega$, and conjugate index $c\tau$ are preserved. No image fallback is used. Running byline/folio, scan-edge material, and the neighboring-page sliver are excluded.

Authority physical 27 · printed 339

Zone: FRENCH MAIN BODY

Printed page 339; French main body. The page develops the Γ -value conjecture through Gauss sums, condition (*), Conjecture 8.11, and the H -equivariant construction of §8.12. The Gauss sum uses $\Psi(x)$ on $k_{\mathfrak{p}}$, and the trace appears only after passage to the residue extension κ'/κ . Products over H -orbits and the SGA $4^{1/2}$ reference are retained as native text. No image fallback is used; running matter and scan-edge slivers are excluded.

Authority physical 28 · printed 340

Zone: FRENCH MAIN BODY

Printed page 340; French main body. The page states Conjecture 8.13 and its (**) variant, including the factor $(2\pi i)^{-k}$, then begins the CM-period comparison of §§8.14–8.15. The basis $e_\sigma \pm e_{c\sigma}$, the Hodge direct factor $(k \otimes E)^+$, and the two-sum formula for $I^+(e_\sigma + e_{c\sigma})$ are retained as native preformatted mathematics. No image fallback is used; copy-matter and scan-edge slivers are excluded.

Authority physical 29 · printed 341

Zone: FRENCH MAIN BODY

Printed page 341; French main body. The page diagonalizes I^+ , states Proposition 8.16, reproduces the commutative comparison diagram of Remark 8.17 natively, identifies $D(\chi)$ with a square root of the discriminant of k_0 , and begins §8.18. The $D_\tau(\chi)$ component, $\bar{S}$ indexing, and quotient $\mathbb{C}^*/\bar{\mathbb{Q}}^*$ are preserved. No image fallback is used; running header/folio and scan-edge material are excluded.

Authority physical 30 · printed 342

Zone: FRENCH MAIN BODY

Printed page 342; French main body. The page gives the period formalism (8.18.1)–(8.18.6), defines η^* , states Proposition 8.20, and reaches equation (8.20.1), ending with the carryover “Par ailleurs,”. The authority has no inverse in $\det_k(1 \otimes y, (k \otimes E)^+)$, and Proposition 8.20 is retained as a product identity. This owned range ends before the actual source-English appendix heading: physical page 31 first continues the French proof and then contains the language boundary. That mixed-page transition is explicitly reserved for P06. No image fallback is used; running matter and scan-edge/neighbor-page slivers are excluded.

Authority physical 31 · printed 343

Zone: SOURCE ENGLISH KOBLITZ OGUS APPENDIX

Printed page 343. The authority page begins by completing the French proof of Proposition 8.20, continues through §8.21 and Remarks 8.22–8.23, and then makes the exact within-page transition to the separately authored source-English “Appendix by N. Koblitz and A. Ogus,” with its title, definitions, H_Q , and opening example. The diplomatic layer preserves the French-to-English boundary; the English reader translates only the French segment and then replays the appendix without retranslation. The appendix authors are not treated as coauthors of Deligne’s main body. No image fallback is used; the left neighboring-page sliver, running matter, folio, and scan-edge band are excluded.

Authority physical 32 · printed 344

Zone: SOURCE ENGLISH KOBLITZ OGUS APPENDIX

Printed page 344; source-English Koblitz–Ogus appendix. The page states the proposition generating H_Q , gives the orbit/stabilizer reduction, the character lemma and its inner-product calculation, and begins the inductive proof. Both source and English readers replay the authority-English text without paraphrase. The right neighboring-page sliver and scan-edge band are excluded; no image fallback is used.

Authority physical 33 · printed 345

Zone: SOURCE ENGLISH KOBLITZ OGUS APPENDIX

Printed page 345; source-English Koblitz–Ogus appendix followed by Bibliographie 1–5. The appendix proof is completed and the algebraicity theorem is replayed with the mandatory authority factors $\Gamma(f) := \pi^{-k} \prod_a \Gamma(\langle a \rangle)^{f(a)}$ and $\Gamma(f)^n = \Gamma(nf)$, satisfying fault gates F05 and F06. The final displayed identity visibly uses a summation sign in the authority and is retained diplomatically rather than silently emended to a product. The left neighboring-page sliver, running matter, folio, and scan-edge band are excluded; no image fallback is used.

Authority physical 34 · printed 346

Zone: BIBLIOGRAPHY TERMINAL ADDRESS MIXED NEUTRAL

Printed page 346; terminal bibliography and address. Entries 6–19, the SGA and SGA 4 / SGA 4^{1/2} references, and “INSTITUT DES HAUTES ÉTUDES SCIENTIFIQUES, BURES SUR YVETTE, FRANCE” are preserved. The blank remainder, running matter, folio, scan-edge band, and the narrow neighboring-article sliver at right are excluded. This is D030 EOF; no image fallback is used.

Mandatory literality and mathematics gates

The independent gate checks the opening Zagier sentence for L_3/L_4 (not L_3/L_5), the exact French phrase “dû à Grothendieck,” diagram (1.3.1), the determinant-line δ passage, $\Gamma(f) := \pi^{-k} \prod_a \Gamma(\langle a \rangle)^{f(a)}$, $\Gamma(f)^n = \Gamma(nf)$, and the six accented section headings. The appendix by N. Koblitz and A. Ogus is attributed only to the appendix; it is not promoted into the main-paper byline.